

Angle sum and difference identities



- 1 a $\sin \alpha \cos \beta + \cos \alpha \sin \beta$ b $\sin \alpha \cos \beta - \cos \alpha \sin \beta$
- 2 $\frac{\tan \theta - \tan x}{1 + \tan \theta \tan x}$
- 3 $\frac{\sqrt{3} \cos \theta + \sin \theta}{2}$
- 4 $\sin 3x \cos y - \cos 3x \sin y$
- 5 $\cos 2x \cos 3y - \sin 2x \sin 3y$
- 6 a $2 - \sqrt{3}$ b $\frac{\sqrt{3} - 1}{2\sqrt{2}}$ c $\frac{1 - \sqrt{3}}{2\sqrt{2}}$ d $\frac{\sqrt{6} + \sqrt{2}}{4}$
e $\frac{\sqrt{2}(\sqrt{3} - 1)}{4}$ f $\sqrt{3} + 2$ g $\frac{\sqrt{6} + \sqrt{2}}{4}$ h $\frac{1}{2}$
i $\sqrt{3}$ j $\frac{\sqrt{3}}{2}$
- 7 a $-\cos x$ b $\sin x$ c $\cos x$ d $-\tan x$
e $\frac{\sqrt{2}(\cos x + \sin x)}{2}$ f $\frac{\cos x - \sqrt{3} \sin x}{2}$ g $\frac{1 + \tan x}{1 - \tan x}$ h $\frac{\sqrt{2}(\cos x + \sin x)}{2}$
- 8 a $\frac{364}{627}$ b $-\frac{644}{333}$
- 9 a $\cos x$ b $-\sin x$ c $\cos x$ d $-\tan x$
- 10 $2 \cos \left(B - \frac{\pi}{2} \right)$
- 11 a $\frac{1}{2}$ b 1 c $\frac{1}{\sqrt{3}}$

12

a $\frac{4}{5}$

b $\frac{8}{17}$



c $-\frac{13}{85}$

d $\frac{84}{85}$

e Quadrant 4

f $-\frac{13}{84}$

13

a $\frac{4}{5}$

b $-\frac{5}{13}$

c $-\frac{63}{65}$

d $\frac{33}{65}$

e $-\frac{63}{16}$

f $-\frac{33}{56}$

g Quadrant 4

h Quadrant 2

14

a i $\frac{\pi}{4} + \frac{\pi}{3}$

ii $\frac{\sqrt{2} - \sqrt{6}}{4}$

b i $\frac{2\pi}{3} + \frac{\pi}{4}$

ii $\frac{\sqrt{6} - \sqrt{2}}{4}$

c i $\frac{\pi}{3} - \frac{\pi}{4}$

ii $2 - \sqrt{3}$

d i $\frac{\pi}{4} - \frac{\pi}{3}$

ii $\sqrt{3} - 2$

15

a $\sin(A + 2B)$

b $\cos(6\theta + x)$

c $\tan 3m$

d $-2 \sin \theta$

16

a $2 \sin(5x) \cos(x)$

b $2 \sin 4x \sin x$

c $2 \cos(x + y) \cos\left(\frac{x - y}{2}\right)$

d $2 \sin 31^\circ \cos 13^\circ$

17 $-\frac{\sqrt{3}}{\sqrt{2}}$

18 $x = \sin \theta, \cos \theta$

19 $x = 20^\circ, 40^\circ, 140^\circ, 160^\circ, 260^\circ, 280^\circ$

Proofs

20 No solution provided.

21

- a No solution provided.
- c No solution provided.
- e No solution provided.
- g No solution provided.



- b No solution provided.
- d No solution provided.
- f No solution provided.
- h No solution provided.

22

- a No solution provided.

- b No solution provided.

23 No solution provided.

Applications

24

a 17 cm

b $17 \cos(\pi x)$

c 2 seconds

25

a 2.3 m

b 2 m

c $-\frac{1}{4} \sin \frac{\pi}{6} x + \frac{\sqrt{3}}{4} \cos \frac{\pi}{6} x + \frac{5}{2}$